

07 Titanic Missing Values

Markdown Cell 1

Experiment 7 — Missing Values Techniques on Titanic Dataset
Source: Titanic PDF

The notebook implements the missing-value techniques shown in the PDF: row deletion, column deletion, arbitrary-value replacement, mean, median, max, min, forward fill, backward fill, and interpolation.

Code Cell 2

```
import pandas as pd
import numpy as np

df=pd.read_csv("titanic_Data_Train.csv")
df.head()
```

Code Cell 3

```
df.shape
df.tail()
```

Code Cell 4

```
df.isnull().sum()
```

Markdown Cell 5

1. Deleting the Missing Values — Entire Row

Code Cell 6

```
# axis=0 means rows; axis=1 means columns
df1=df.dropna(axis=0)
print(df1.shape)
print(df1.isnull().sum())
```

Markdown Cell 7

2. Deleting the Missing Values — Entire Column

Code Cell 8

```
df2=df.drop(['Age', 'Cabin', 'Embarked'], axis=1)
print(df2.isnull().sum())
```

Markdown Cell 9

3. Imputing Missing Values — Arbitrary Value

Code Cell 10

```
df3=df.copy()
df3['Age']=df3['Age'].fillna(0)
df3['Cabin']=df3['Cabin'].fillna(0)
df3['Embarked']=df3['Embarked'].fillna(0)
print(df3.isnull().sum())
```

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4. Imputing Missing Values — Mean

Code Cell 12

```
df4=df.copy()
df4['Age']=df4['Age'].fillna(df4['Age'].mean())
print(df4.isnull().sum())
```

Markdown Cell 13

5. Imputing Missing Values — Median

Code Cell 14

```
df6=df.copy()
df6['Age']=df6['Age'].fillna(df6['Age'].median())
print(df6.isnull().sum())
```

Markdown Cell 15

6. Imputing Missing Values — Maximum

Code Cell 16

```
df7=df.copy()
df7['Age']=df7['Age'].fillna(df7['Age'].max())
print(df7.isnull().sum())
```

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7. Imputing Missing Values — Minimum

Code Cell 18

```
df8=df.copy()
df8['Age']=df8['Age'].fillna(df8['Age'].min())
print(df8.isnull().sum())
```

Markdown Cell 19

8. Forward Fill

Code Cell 20

```
df9=df.copy()
df9['Cabin'] = df9['Cabin'].ffill()
print(df9.isnull().sum())
```

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9. Backward Fill

Code Cell 22

```
df10=df.copy()
df10['Cabin'] = df10['Cabin'].bfill()
print(df10.isnull().sum())
```

Markdown Cell 23

10. Interpolation

Code Cell 24

```
df11=df.copy()
df11['Age'] = df11['Age'].interpolate()
print(df11.isnull().sum())
```